03-621 Week 9

Advanced Quantative Genetics

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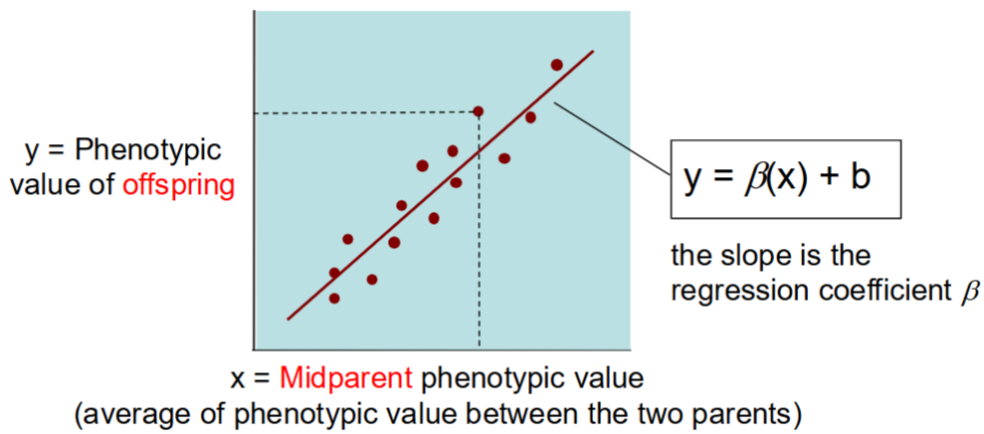
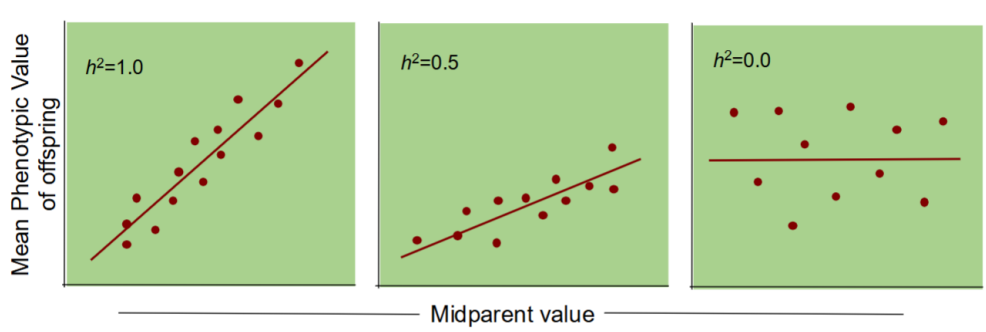
h^2 from Parent-Offspring Regression



$$
\begin{aligned}
\beta &= \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)} \\
&= \frac{\sum(x_i-\bar{x})(y_i-\bar{y})/(N-1)}{\sum(x_i-\bar{x})^2/(N-1)} \\
&= \frac{(1/2)V_A}{(1/2)V_p} \\
&= \frac{V_A}{V_p} \\
&= h^2
\end{aligned}
$$

Possible values of h^2 and their interpretation



- $h^2 = 1.0$: Genes with **additive** effects determine **all** of the differences in phenotypic value
- $1.0 > h^2 > 0$: Genes with additive effects, dominance, and/or epistasis may contribute, as well as the environment
- $h^2 = 0.0$: **No additive genetic contribution** to the differences in phenotypic value

h^2 for Some Human Disease Traits

Trait	h^2
Cleft lip	65
Spina bifida	65
Club foot	80
Pyloric stenosis	85
Dislocation of hip	95
Hydrocephalus	40
Bipolar affective disorder	∼100
Multiple sclerosis	∼100
Schizophrenia	∼100
Diabetes	80
Epilepsy	40

These particular examples are all **threshold traits:** the heritability is for the quantitative phenotype, which is the **midparent value for the underlying liability (risk)**.

Note that h^2 for a given trait can vary:

- between different populations (different gene pools)
- between different locations (different environments)
- between different generations of the same population
  - Before / After selection
  - With changes in environment (e.g., diet)

For the same trait and environment:

- h^2 will be higher in a population with high genetic diversity
- h^2 will be lower in a population with low genetic diversity

Selection

h^2 decreases when strong selection pressure reduces genetic diversity.

- Consider a trait with phenotype values 4 to 16, with the (normal distribution) centered at 10.
- If suddenly selection pressure favors only those with above phenotype value 12, then the genotypes contributing to the next generation would be those with high phenotype value
- This reduces the gene pool and therefore h^2 decreases

Can we predict *how* much the average phenotype will actually chnage under selection?
Can we predict the *average* phenotype of progeny for any two parents?

- Yes and Yes!

If we know the h^2 for a trait in a population, the equation for parent-offspring regression allows us to find the **average** phenotype of progeny:

$$y = h^2 x + b$$

- y = expected offspring value
- x = mid-parental value

We can also obtain h^2 from (and apply it to)

- Group Selection Experiments
- Threshold Traits

To do this we modify the regression equation to obtain the "Breeder's Equation'

Linking h^2 and Selection: The Breeder's Equation

- Parent-offspring regression: $y = h^2x + b$
- Rewrite in terms of differences from the original population M: $(y - M) = h^2(x - M)$
- Average this over the selected parents and their offspring: $(M' - M) = h^2(M^* - M)$
  - M' = mean of progeny
  - M^* = mean of selected parents
- **The Breeder's Equation:** $R = h^2S$
  - $R = (M' - M)$. "**R**esponse to selection"
  - $S = (M^* - M)$. "**S**election differential"
- The amount of evolutionary change (R) is determined by the narrow-sense heritability (h^2) and the strength of selection (S).
- h^2 can be **obtained** from selective breeding experiments *or* from the incidence of a threshold trait among children of affected parents in a population

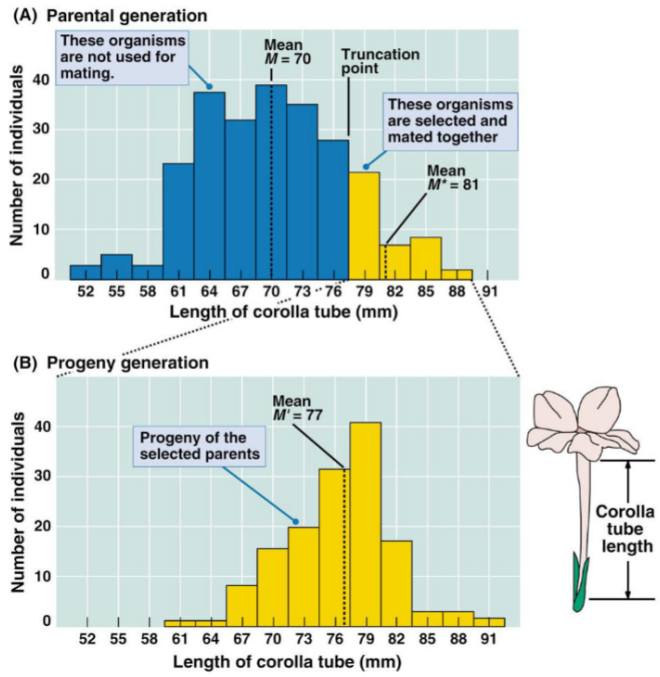
$$h^2 = \frac{R}{S} = \frac{M' - M}{M^* - M}$$

- h^2 can be used to **predict** the phenotypic mean of a progeny population obtained by selective breeding (or natural selection)

$$M' = M + h^2(M^* - M)$$

Example: Effect of Selection on Size of Tobacco Flowers

We can obtain h^2 from selection experiments:



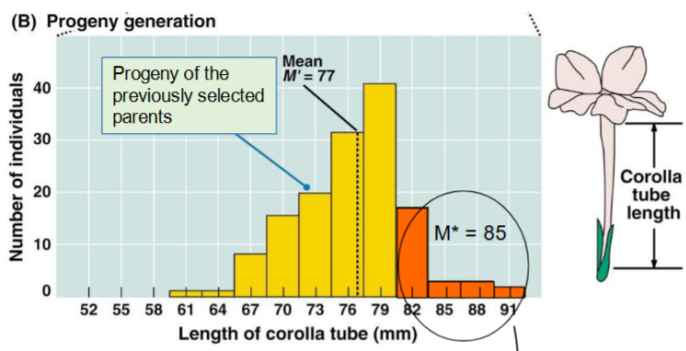
$$R = h^2 S$$

$$h^2 = \frac{R}{S}$$

$$h^2 = \frac{M' - M}{M^* - M}$$

$$h^2 = \frac{77-70}{81-70} = 0.636$$

And we can predict the effect of selection:



If parents with M*= 85 mm are selected from Generation 2, what will be the mean flower length of Generation 3?

$$h^2 = \frac{M' - M}{M^* - M}$$

$$M' = M + h^2 (M^* - M)$$

$$M' = 77 + (0.636)(85-77) = 82 \text{ mm}$$

And so on…

For how long? Response over multiple generations

- **strictly speaking**, an initial h^2 from the breeders' equation only holds for predicting a **single** generation of response.
  - Reason: allele and genotype frequencies change after selection so that $\sigma^2_A, \sigma^2_D, \sigma^2_I$, and σ^2_{gxe} can change. Environment can also change.
- In practice, an initial h^2 usually does well for 5-10 generations, sometimes much longer

This depends on:

1. Degree of environmental change between generations
2. Degree of genetic change between the generation in which h^2 was estimated and the generation in which selection is applied. Strong/Fast vs. Moderate/Gradual selection.

h^2 does not always decrease after selection

Example: Oil content in corn

- Original h^2 for oil content in corn has held since 1896, or 125 generations(!)
- Implies gradual selection has not exhausted genetic diversity

Example: Lactation milk yield in dairy cows

- h^2 for lactation milk yield in dairy cattle has actually *increased* over time:
  - $\sim 25\%$ in the 1970s
  - $\sim 40\%$ currently
- The reason for this is due to improvements in feeding and veterinary care.
- Leads to reduced V_E

Obtaining h^2 values for human traits

1. Continuous and Meristic traits
   - Phenotypic values can be measured or counted directly
     - Use parent-offspring regression
2. Threshold traits
   - We can't measure the underlying liability directly.
   - We can't do selection experiments
   - However, we can use the Breeders' Equation, adapted to incidence distributions

The Breeder's Equation for Selection

Our examples so far have represented **Truncation Selection**

- e.g., only individuals **above** a certain phenotypic value (T) contribute to the next generation.

Expression of Threshold Traits is equivalent to truncation selection!

- The quantitative phenotype with a continuous distribution is the underlying **liability**. The trait is expressed if liability **exceeds** a threshold.
- This is the same situation as threshold selection. (Members below the threshold do not reproduce.)
- This equivalence gives us a way of estimating h^2 for the liability of threshold traits using the Breeder's Equation

What Proportion of the *Liability* is Inherited?

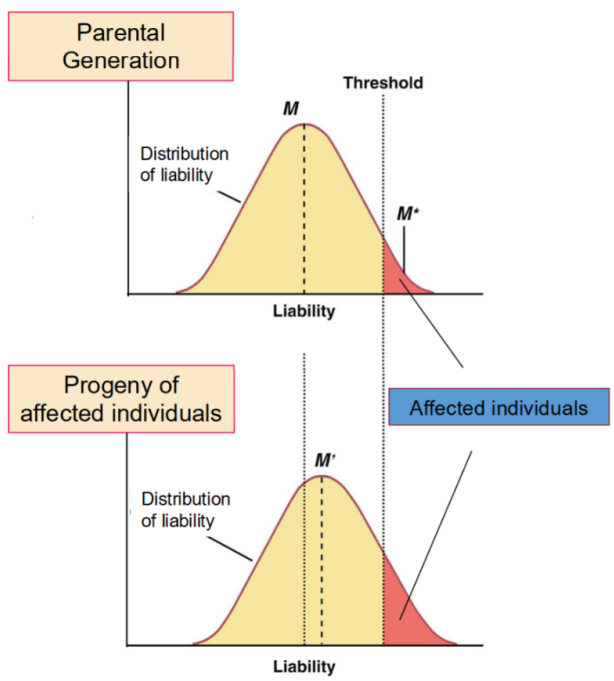
Recall:

$$h^2 = \frac{M' - M}{M^* - M} = \frac{R}{S}$$

- M = mean liability of the base population
- M^* = mean liability of affected individuals
- M' = mean liability of progeny from affected individuals

How do we obtain M, M^*, and M'?

- We don't need them: only the differences (R, S), which can be calculated from the proportions of affected individuals, using properties of the *standard normal distribution*.



This determines what proportion of the progeny of affected individuals in the population will be above threshold and develop the trait.

- Note: A **standard normal distribution** is one with mean $\mu = 0$ and variance $\sigma^2 = 1$.
- Any arbitrary normal distribution can be converted into a "standard" normal distribution by expressing the ordinate values in terms of standard deviations from the mean:

$$z = (x - \mu)/\sigma$$

Standardizing the Breeder's Equation

We express $R = M' - M$ and $S = M^* - M$ in terms of standard deviations of the mean.

- Let $i = (M^* - M)/\sigma$, so $S = i\sigma$. (i stands for "selection intensity")
- Let $x_1 = (Threshold - M)/\sigma$ and $x_2 = (Threshold - M')/\sigma'$, so $R = (x_1 - x_2)\sigma$. ($\sigma' \sim \sigma$)

Now, we get $h^2 = \frac{R}{S} = \frac{(x_1 - x_2)\sigma}{i\sigma}$

$$h^2 = \frac{x_1 - x_2}{i}$$

Now we're in business, because for a standard normal distribution,

$$P(x) = \sqrt{\frac{1}{2\pi}} \exp\left(-\frac{x^2}{2}\right)$$

The area under the curve from $-\infty$ to X is

$$\Phi(X) = \int_{-\infty}^{X} P(x)\,\mathrm{d}x$$

or, the "cumulative distribution function". Similarly, the area unde the curve from X to ∞ would be $1-\Phi(x)$.

- The **mean value** for individuals within area $1 - \Phi(X)$ is

$$\mu^* = \frac{\exp\left(-\frac{x^2}{2}\right)}{\sqrt{2\pi}(1 - \Phi(x))}$$

Now we have what we need to obtain h^2 for threshold traits:

$$h^2 = \frac{x_1 - x_2}{i}$$

1. If x is the distance between Threshold and Mean:
   - $1 - \Phi(x)$ is the proportion of the population that exceeds the threshold and is **affected**
2. Given $1 - \Phi(x_1)$ and $1 - \Phi(x_2)$, we can obtain x_1 and x_2 from the distribution function $\Phi(x)$
3. Then, we can calculate i:

$$i = \mu^* = \frac{\exp\left(-\frac{x_1^2}{2}\right)}{\sqrt{2\pi}(1 - \Phi(x_1))}$$

Note: in practice, if we want to find $\Phi(x)$, we would use a statistical table, or use a calculator.

h^2 for Threshold Traits . . . again

$$h^2 = \frac{x_1 - x_2}{i}$$

Re-emphasize that:

- x_1 is the number of standard deviations by which the Threshold exceeds the mean liability in the **base (i.e., "general") population**.
- x_2 is the number of standard deviations by which the Threshold exceeds the mean liability among **offspring of affected** individuals.
- i is the number of standard deviations by which μ^* exceeds the mean liability in the **base population**.

Modification for Estimation for Rare

Threshold Traits The h^2 equation above holds for *common* traits where x_2 is calculated from progeny of **two affected parents** (so midparent liability is $\mu^* = i$)

- **If the trait is rare**, only 1 parent is likely affected and has average liability $\mu^* = i$
- The other parent comes from the general population and has average liability 0.
- Therefore, **midparent liability** is $(i + 0)/2 = i/2$.

$$\therefore h^2 = \frac{x_1 - x_2}{i/2} = \frac{2(x_1 - x_2)}{i}$$

Example: Calculating h^2 for Threshold Traits

For this example, consider Rheumatic Fever. The frequency in the general population is $0.005 = 1 - \Phi(x_1)$.

- So, $\Phi(x_1) = 0.995$
- From the distribution function we find: when $\Phi(x_1) = 0.995 \rightarrow x_1 = 2.57$. So, $i = 2.8988$.

Frequency among offspring of affeted individuals is $0.05 = 1 - \Phi(x_2)$.

- So, $\Phi(x_2) = 0.95$
- From the distribution function we find: when $\Phi(x_2) = 0.95 \rightarrow x_2 = 1.64$.

Therefore,

$$h^2 = \frac{x_1 - x_2}{i/2} = \frac{2.57 - 1.64}{2.8988/2} = 0.64$$

So, 64% of the liability is heritable.

- Since $H^2 \geq h^2$, **at least** 64% of the variance in liability is due to genotype.

Remember: Broad- versus Narrow-Sense Heritability

$$H^2 = \frac{\sigma_g^2}{\sigma_p^2}$$

Includes:

- **Additive** effects of alleles (σ_A^2)
- **Dominance** effects (σ_D^2)
- **Epistasis** effects (σ_I^2)
- $\sigma_g^2 = \sigma_A^2 + \sigma_D^2 + \sigma_I^2$

$$h^2 = \frac{\sigma_A^2}{\sigma_p^2}$$

Includes:

- Only the **additive** effects of alleles
- E.g., only the **additive** effects lead to heritable differences in phenotypic mean

h^2 is the proportion of phenotypic variance that can be used to predict the changes in the phenotypic mean after **selection on a population** or among **progeny of individuals.**

- Generally, $h^2 < H^2$.
- $h^2 = H^2$ when **all** alleles affecting a trait are **additive** in their effects (there is no dominance or epistasis)

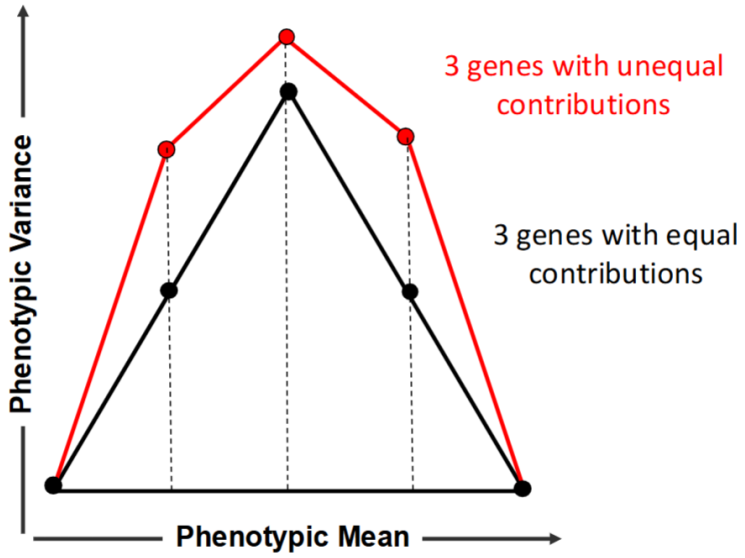
Review: Variance and Gene Number

$$n = \frac{D^2}{8V_g}$$

The following phenomena increase when V_g is increased, leading to underestimates of gene number (n):

- Dominance
- Epistasis (examples: masking; multiplicative)
- Unequal gene/allele contribution
- Linkage

Effect of Unequal Contributions or Partially Linked Genes



Partial linkage has similar effects because recombination in F1 germlines generates recombinant chromosomes with different contributions.